

Experimental Design: Evaluating the Reliability and Security of LLM-Based Tools in Software Development

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1 Problem Statement

The integration of Large Language Models (LLMs) into the software development lifecycle promises significant productivity gains. However, the reliability and security of the generated artifacts remain critical concerns for industrial adoption. This study evaluates the research question: To what extent can developers rely on specific LLM-based tools for code generation and security-focused code review?

2 Experimental Tasks

The study focuses on two fundamental coding-related tasks:

- **Code Generation:** Evaluating the functional correctness and compilation success of algorithmic implementations.
- **Code Review (Security Vulnerabilities):** Assessing the model's ability to identify memory safety issues and injection vulnerabilities, prioritizing security over general bug detection.

3 Experimental Design and Variables

Following a structured experimental protocol, the study isolates specific factors to ensure scientific validity.

3.1 Independent Variables

- **LLM Model:** ChatGPT 5.2 vs. Claude Sonnet 4.6.
- **Programming Language:** Python (High-level, interpreted) vs. C (Low-level, manual memory management).

3.2 Dependent Variables

- **Execution Success Rate:** The percentage of generated code snippets that execute correctly.
- **Vulnerability Recall:** The percentage of successfully identified security vulnerabilities in a curated dataset.

3.3 Control Variable

- **Development Environment:** All interactions are conducted via the Cursor IDE to maintain environmental consistency.

4 Methodology and Data Collection

A custom dataset will be curated by adapting code from real-world repositories and security benchmarks.

- **Sampling:** To ensure statistical significance for binomial variables, each experimental cell will include a sample size substantially larger than $N = 10$.
- **Reference and Adaptation:** All reused code snippets will be properly referenced, and any modifications for the experimental scenario will be documented.

5 AI Disclosure

This document was reviewed and refined with the assistance of ChatGPT and similar tools, which helped check grammar, correct typos, and enhance clarity. Any AI-generated text or content is clearly indicated within the document and is used only to a limited extent. The overall content and ideas remain solely the responsibility of the authors.

References

1. DEI-FCTUC. (2026). *MEI Assignment - Experiment design: Experimental Methods in Computer Science*. Version 1.0.